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Friction plate having a groove pattern formed by means of friction lining pads

Abstract

A frictional surface for an annular wet friction system with external lubrication includes a groove pattern with first pentagonal friction lining pads, second pentagonal frictional lining pads, and a groove. The first pentagonal friction lining pads are arranged radially outwards in a first row and the second pentagonal friction lining pads arranged radially inwards in a second row. The groove extends circumferentially between the first pentagonal friction lining pads and the second pentagonal friction lining pads in a zigzag or undulating manner. In an embodiment, each of the first pentagonal friction lining pads has a triangular geometry with a first tip directed radially inwards and each of the second pentagonal friction lining pads has a triangular geometry with a second tip directed radially outwards.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5669474	12/1996	Dehrmann et al.	N/A	N/A
5878860	12/1998	Pavangat et al.	N/A	N/A
8205734	12/2011	Sudau et al.	N/A	N/A
8474590	12/2012	Fabricius et al.	N/A	N/A
10502269	12/2018	Takakura et al.	N/A	N/A
11067133	12/2020	Tepper et al.	N/A	N/A
2012/0118696	12/2011	Fabricius	192/107R	F16D 13/72
2012/0175216	12/2011	Hiramatsu et al.	N/A	N/A
2017/0254368	12/2016	Hartner	N/A	F16D 13/683
2017/0363154	12/2016	Heitzenrater et al.	N/A	N/A
2018/0216674	12/2017	Takakura	N/A	F16D 13/74
2018/0328415	12/2017	Langenkaemper et al.	N/A	N/A
2018/0372166	12/2017	Carey et al.	N/A	N/A
2019/0345988	12/2018	Dannwolf et al.	N/A	N/A
2020/0049206	12/2019	Tepper	N/A	F16D 65/128
2020/0393004	12/2019	Tepper	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102018105214	12/2018	DE	N/A
102018105215	12/2018	DE	N/A
2028382	12/2013	EP	N/A

3374652	12/2019	EP	N/A
2004211781	12/2003	JP	N/A
WO-2016180540	12/2015	WO	F16D 13/648
2019/120370	12/2018	WO	N/A
2022033631	12/2021	WO	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the United States National Phase of PCT Appln. No. PCT/DE2022/100348 filed May 5, 2022, which claims priority to German Application Nos. DE102021114508.2 filed Jun. 7, 2021 and DE102021120275.2 filed Aug. 4, 2021, the entire disclosures of which are incorporated by reference herein.

TECHNICAL FIELD

(2) The present disclosure relates to a wet multiple disc brake with external oiling.

BACKGROUND

(3) Wet multiple disc clutches and brakes are widely used in conventional power-shiftable transmissions, in hybrid modules in heavy-duty drive trains or in shiftable e-axles, and they represent high-performance, heavy-duty components. The demands for lower CO_2 emissions and improved efficiency of drive trains in automotive applications are of great importance. In addition to the reduction of load-independent losses in shifting elements, the thermal load and sufficient cooling must be considered. The groove pattern of the friction disc plays a central role in the trade-off between friction characteristics, thermal balance, and efficiency.

(4) WO 2019/120 370 A1 as well as U.S. Pat. No. 8,474,590 B2 and EP 3 374 652 B1 each disclose annular wet friction parts with grooves in the frictional surface.

(5) In the case of annular wet friction systems with external lubrication—also referred to as external oiling in the context of this publication—and in particular in the case of multiple disc brakes with external oiling (cf. FIG. **2**), proven groove patterns (as for internal oiling) cannot be used.

SUMMARY

(6) The present disclosure is directed to improving the convective cooling/cooling effect and minimizing drag losses in wet friction systems with external lubrication, for example in multiple disc brakes with external oiling, by means of a suitable groove pattern.

(7) The present disclosure provides a groove pattern for an annular wet friction system with external lubrication. For example, the annular wet friction system is a wet multiple disc brake with external oiling.

(8) The groove pattern according to the disclosure for an annular wet friction system with external lubrication thus provides that the frictional surface has a circumferential groove extending across the circumference in a zigzag or undulating manner.

(9) In wet friction systems with external lubrication, such a groove pattern improves the cooling effect and reduces drag losses.

(10) In an exemplary embodiment of the groove pattern, the circumferential groove extending across the circumference in a zigzag or undulating manner is arranged, in the radial direction, between first pentagonal friction lining pads arranged radially outwards in a first row of pads and

second pentagonal friction lining pads arranged radially inwards in a second row of pads. The first and second pentagonal friction lining pads create a double row groove pattern with a circumferential groove extending across the circumference in a zigzag or undulating manner. The circumferential groove extending across the circumference in a zigzag or undulating manner may be arranged centrally between the first pentagonal friction lining pads and the second pentagonal friction lining pads. The first pentagonal friction lining pads may be substantially of the same design. The same applies to the second pentagonal friction lining pads.

(11) In a further exemplary embodiment of the groove pattern, the first and second pentagonal friction lining pads have a rectangular geometry with an immediately adjacent triangular geometry, wherein tips of the triangular geometry of the first pentagonal friction lining pads are directed radially inwards and tips of the triangular geometry of the second pentagonal friction lining pads are directed radially outwards. The size and shape of the triangular geometry of the first and second pentagonal friction lining pads allows the shape and size of the circumferential groove extending across the circumference to be varied and adapted to a desired requirement profile.

(12) In a further exemplary embodiment of the groove pattern, first radial grooves are arranged in the circumferential direction between the first pentagonal friction lining pads, in each case, and second radial grooves are arranged in the circumferential direction between the second pentagonal friction lining pads, in each case. The tips of the triangular geometry of the first pentagonal friction lining pads face the second radial grooves, and the tips of the triangular geometry of the second pentagonal friction lining pads face the first radial grooves. The first radial grooves and the second radial grooves each open into the circumferential groove extending across the circumference in a zigzag or undulating manner.

(13) In a further exemplary embodiment of the groove pattern, the first pentagonal friction lining pads have a V-shaped double groove. A V-shaped double groove means, with regard to the first pentagonal friction lining pads, that they have two grooves, each of which is arranged in a V-shape. The two grooves arranged in a V-shape are spaced apart from one another in the circumferential direction, and the distance between the grooves of the double groove decreases radially outwards. A groove angle enclosed by the two grooves of the double groove may be between twenty and thirty degrees, e.g., 24.3 degrees. The two grooves of the V-shaped double groove can be designed as embossed grooves. The two grooves of the V-shaped double groove can also be designed as segmentation grooves. If required, one of the grooves of the V-shaped double groove can also be designed as an embossed groove, while the other of the two grooves of the V-shaped double groove is designed as a segmentation groove.

(14) In a further exemplary embodiment of the groove pattern, at least two second pentagonal friction lining pads are integrally connected to each other and are divided only by an embossed groove. According to an exemplary embodiment, two second pentagonal friction lining pads are integrally connected to each other in each case. According to a further exemplary embodiment, three second pentagonal friction lining pads are integrally connected to each other. According to a further exemplary embodiment, all second pentagonal friction lining pads are integrally connected to each other. The second pentagonal friction lining pads connected to each other are divided by an embossed groove in each case.

(15) In a further exemplary embodiment of the groove pattern, pad inner angles in pad corners of the first and second pentagonal friction lining pads are between ninety and one hundred and fifty degrees. This angular range has proven to be advantageous with regard to the desired effect in the operation of the groove pattern.

(16) In a further exemplary embodiment of the groove pattern, all pad corners are rounded along their circumferential contour. The rounding radii may be greater than or equal to one millimeter.

(17) In a further exemplary embodiment of the groove pattern, the first and second friction lining pads have widths and heights that have a width to height ratio that is greater than one and less than three. The width to height ratio of the first pentagonal friction lining pads may be 2.58. The width

to height ratio of the second pentagonal friction lining pads may be 2.33.
(18) In a further exemplary embodiment of the groove pattern, a radial flow cross-section between the first pentagonal friction lining pads is larger than a radial flow cross-section between the second pentagonal friction lining pads. The radial flow cross-section is defined by the size of the radial grooves between the first and second pentagonal friction lining pads, respectively. In this regard, the radial grooves can be segmentation grooves as well as embossed grooves.
(19) The present disclosure further relates to a friction lining pad for a groove pattern described above. The friction lining pads can be traded separately.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Further advantages and advantageous configurations of the present disclosure are the subject of the following figures and the description thereof.
(2) In the figures:
(3) FIG. **1** shows various known patterns of common grooving of friction linings:
(4) FIGS. **2***a* and **2***b* show a wet multiple disc brake with external oiling:
(5) FIG. **3***a* shows a schematic temperature curve of a friction disc with external oiling:
(6) FIG. **3***b* shows oil routing and cooling in a closed state of the friction disc with external oiling:
(7) FIGS. **4***a*-**4***c* show drag torques of multiple disc clutches/brakes:
(8) FIG. **4***d* shows oil removal in an open state of the friction disc with external oiling:
(9) FIGS. **5***a* and **5***b* show groove patterns for friction systems with external lubrication;
(10) FIGS. **6***a*-**6***i* show groove patterns for friction systems with external lubrication:
(11) FIGS. **7***a*-**7***c* show groove patterns for friction systems with external lubrication:
(12) FIGS. **8***a*-**8***f* show groove patterns for friction systems with external lubrication:
(13) FIGS. **9***a*-**9***d* show groove patterns for friction systems with external lubrication:
(14) FIG. **10***a* shows schematic cooling oil flow in the closed state of a groove pattern for friction systems with external lubrication: and
(15) FIG. **10***b* shows schematic oil removal in the open/closing state of a groove pattern for friction systems with external lubrication.
DETAILED DESCRIPTION
(16) Various known groove patterns **62** to **69** are shown in plan view in FIG. **1**. A friction disc with a frictional surface not equipped with grooves is designated with **61**. Radially on the inside, the friction disc has an internal toothing for hooking the friction disc into a multiple disc carrier (not shown).
(17) The groove pattern **62** comprises radial grooves. The groove pattern **63** comprises cross grooves. The groove pattern **64** comprises parallel grooves arranged in groups. The groove pattern **65** comprises blind grooves arranged crosswise. The groove pattern **66** comprises spiral grooves. The groove pattern **67** comprises intersecting grooves. The groove pattern **68** comprises sunburst grooves. The groove pattern **69** comprises an annular groove with pressure relief holes.
(18) The groove patterns are used to cool the discs with a flow of oil, even when the shifting element is closed. In addition, the grooves serve to cut the oil film and thereby stabilize the friction coefficient. In this way, a desired friction behavior is created in a shift. When the shifting element is in the open state, the drag torque can be influenced and reduced by the grooves.
(19) In FIGS. **2***a* and **2***b*, a wet multiple disc brake **20** is shown schematically in different views. FIG. **2***a* shows various lubrication systems **21**, **22** and **23** of wet multiple disc clutches or multiple disc brakes. The lubrication systems **21** to **23** can be implemented differently for wet multiple disc clutches and brakes, depending on the application.
(20) In general, the cooling oil of the friction systems is supplied from the inside either actively, for

example in the case of double clutches with pressure oiling, or passively, for example in shifting elements in stepped automatic transmissions with passive oil distribution in the transmission, as illustrated by an arrow **24** and a double arrow **25**. Depending on the design of the transmission, the friction system can also be operated in an oil bath, as suggested for **23**. In the special case of multiple disc brakes, such as those used in stepped automatic transmissions, hybrid transmissions or e-axles, active oiling from the outside can be useful, as indicated by an arrow **26** for **22**.

(21) An arrow in FIG. **2***b* indicates that an inner multiple disc carrier **27** of the wet multiple disc brake **20** rotates at a speed ω. One of a total of four friction discs **28** is suspended in the inner multiple disc carrier **27**. The friction discs **28** are connected to the inner multiple disc carrier **27** in a non-rotatable manner by means of a corresponding internal toothing.

(22) The friction discs **28** are each arranged axially between two steel discs **29** which are connected in a non-rotatable manner to an outer multiple disc carrier **30** of the wet multiple disc brake **20**. The arrows r.sub.i and r.sub.a indicate an inner radius and an outer radius of annular disc-like frictional surfaces between the steel discs **29** and the friction discs **28** when the wet multiple disc brake **20** is closed. An arrow h in FIG. **2***b* shows that the steel discs **29** are spaced apart from the friction discs **28** in the axial direction when the multiple disc brake **20** is in the open state. The term “axial” refers to a rotational axis **33** of the wet multiple disc clutch **20**.

(23) Discs brakes are generally used as internal shifting elements for shifting under load in planetary gear transmissions. Wet discs brakes **20**, as shown in FIGS. **2***a* and **2***b*, are used in automatic transmissions, DHT transmissions and/or in multi-speed e-axles.

(24) FIG. **3***b* shows the wet multiple disc brake **20** in a plan view on a friction disc **28**. In a circle **26**, arrows indicate an oil supply from the outside for cooling the multiple disc brake **20** in the closed state. In a circle **36**, it is indicated that a suitable groove pattern is intended to direct a cooling oil flow along the circumference of the friction ring to thereby provide complete, uniform as well as effective convective cooling of the friction system after a shifting event. An exit of the cooling oil is indicated by a circle **37**. The cooling oil flow should exit at the lowest point of the friction system, as far as possible. A premature outflow of the cooling oil on the inner diameter at the oil entry point and/or along the circumference on the outer diameter should be prevented or minimized.

(25) The friction disc **28** is equipped with a frictional surface **34** and an internal toothing **35**. A desired groove pattern is provided in the frictional surface **34**.

(26) A Cartesian coordinate diagram with an X-axis **31** and a Y-axis **32** is shown in FIG. **3***a*. A time in a suitable time unit is plotted on the X-axis **31**. A temperature or a speed is plotted on the Y-axis **32** in a suitable unit in each case. In a rectangle **40**, the multiple disc brake is closed. To the right of the rectangle **40**, the multiple disc brake is open. **38** illustrates a speed drop on closing of the multiple disc brake. **39** illustrates a speed increase on opening of the multiple disc brake. In an ellipse **44**, non-uniform temperature distributions can be seen on the circumference of the multiple disc brake due to a non-uniform cooling oil distribution. When the brake is closed, the friction discs and the steel discs in the disc pack of the multiple disc brake are pressed together.

(27) A Cartesian coordinate diagram with an X-axis **41** and a Y-axis **42** is shown in FIG. **4***a*. A speed difference is plotted on the X-axis **41** in a suitable speed unit. A drag torque is plotted on the Y-axis **42** in a suitable unit. A curve **43** shows a drag torque curve in different sections **45**, **46** and **47**. There is a linear course **70** up to a point **71**. After a maximum **72** there is a drop **73** in the drag torque curve. A dotted line indicates relative movements, in particular wobbling movements of the discs, which lead to a renewed increase in the drag torque.

(28) FIG. **4***b* shows a shear flow of oil between a friction disc **28** and a steel disc **29**. A shift **50** of an air intake to low speeds is indicated in FIG. **4***c*. A suitable groove pattern is intended to improve the oil removal of the brake and thus the drag losses.

(29) A circle **48** in FIG. **4***d* indicates that oiling when the multiple disc brake **20** is in the open state should be reduced or minimized as far as possible, namely advantageously by means of a suitable

groove pattern. In a circle **49**, the oil removal is indicated by an arrow. When the multiple disc brake **20** is in the open state, rapid oil removal/spinning free is desirable. Both the separation of the discs and the oil removal can be assisted by the groove pattern.
(30) Cooling in the Closed State (No Rotation) (FIG. **3**, FIG. **10**)
(31) The design of the groove pattern facilitates the cooling oil supply from the outside by means of a low flow resistance, and a targeted oil flow minimizes early outflow of the cooling oil from the friction system on the one hand and enables uniform cooling over the circumference of the friction system on the other (improvement of convective cooling). This can improve the thermal balance of the shifting element and reduce cooling times.
(32) Drag Losses in the Open State (FIG. **4**, FIG. **10**)
(33) By taking into account the interdependencies of the air intake/separation behavior and their effect on drag losses, the design of the groove pattern (influencing the pressure level/distribution in the lubrication gap) can minimize drag losses. At the same time, additional passive oiling of the friction system from the inside of the transmission is reduced. This supports the goal of a low-loss multiple disc brake as a shifting and separating element for hybrid modules and e-axles.
(34) Functional Description of Groove Patterns for Friction Systems with External Lubrication (FIGS. **5***a*-**5***b*)
(35) Drag Losses—Open State
(36) Groove cross-sectional area increasing from inside to outside—diffuser effect. Improved oil removal in the open state of the brake (rotation of the friction discs). This creates an additional pressure reduction in the groove/lubrication gap between the discs and thus shifts the air intake to low speeds. The drag torque can thus be reduced.
(37) Indirect oiling from the inside is reduced due to the reduced flow cross-section at the inner diameter. This can additionally support the oil removal of the friction system in the open state and reduce drag losses.
(38) Friction Value Build-Up—Closing State
(39) Improved friction value build-up due to effective oil removal in the lubrication gap through optimized over-embossing of the outer row of pads (Pad1). Aggregated enlarged pads of the inner row of pads are also over-embossed for improved oil removal.
(40) Cooling—Closed State
(41) Wide groove channels **121** of the outer row of pads promote oil supply from the outside in the closed state of the brake. Curved zigzag groove **120** centered for distribution of cooling oil over the circumference of the friction system. The reduced flow cross-section on the inner ring **122**, e.g., by combined pads **112** as in FIG. **5***a*, reduces the flow of cooling oil out of the friction contact. The cooling oil supplied from the outside can be optimally guided to the surface of the counter friction disc by means of a coordinated over-embossing **125**. This leads to improved distribution of the cooling oil and increases the contact area for convective heat transfer between the cooling oil and the steel disc.
(42) FIGS. **5** through **10** illustrate a groove pattern **115** in various embodiments. The groove pattern **115** comprises first friction lining pads **111** and second friction lining pads **112**. The friction lining pads **111**, **112** are arranged on a carrier disc **113**. The carrier disc **113** with the friction lining pads **111**, **112** is referred to as the friction disc.
(43) The first and second pentagonal friction lining pads **111**, **112** each have a rectangular geometry with an immediately adjacent triangular geometry. Peaks of the triangular geometry of the first pentagonal friction lining pads **111** are directed radially inwards. Peaks of the triangular geometry of the second pentagonal friction lining pads **112** are directed radially outwards.
(44) A circumferential groove **120** extends radially in the circumferential direction between the first friction lining pads **131** to **134** and the second friction lining pads **141** to **145**. The arrangement and shape of the first pentagonal friction lining pads **111** and the second pentagonal friction lining pads **112** results in a zigzag or undulating course of the circumferential groove **120**.

(45) A first radial groove **121** is arranged between two adjacent first friction lining pads **131**, **132** in each case. A second radial groove **122** is arranged between two adjacent second friction lining pads **141**, **142** in each case.
(46) The first pentagonal friction lining pads **111** are provided with a V-shaped double groove **123**, as can be seen in FIG. **5***a* using the example of the first friction lining pad **132**. The V-shaped double groove **123** comprises two V-shaped embossed grooves **124** and **125** in the first friction lining pad **132**.
(47) FIG. **5***a* further shows that two second friction lining pads **142**, **143** and **144**, **145** are integrally connected to each other in each case. The integrally connected second friction lining pads **142**, **143** and **144**, **145** are divided by only one embossed groove **126** in each case. The embossed groove **126** replaces a radial groove **127** designed as a segmentation groove shown in FIG. **5***b*.
(48) FIG. **6***a* shows that the first pentagonal friction lining pads **111** and the second pentagonal friction lining pads **112** can also be designed without embossed grooves. In FIG. **6***b*, pad inner angles **1** are drawn in the first friction lining pad **136**. The pad inner angles **1** may be between ninety and one hundred and fifty degrees. FIG. **6***c* illustrates that pad outer edges are provided with rounding radii **2**, which may be greater than or equal to one millimeter.
(49) In FIG. **6***d*, a double arrow **3** indicates a width of the first friction lining pad **136**. A height **4** of the first friction lining pad **136** is indicated by a double arrow **4**. A width **3** to height **4** ratio of the first pentagonal friction lining pads **111** is between one and three, e.g., **2**.**58**.
(50) In FIG. **6***e*, double arrows **5** indicate an outer flow cross-section between the first pentagonal friction lining pads **111**. The outer flow cross-section **5** is larger than an inner flow cross-section **6** indicated by a double arrow between two second pentagonal friction lining pads **112**.
(51) In FIG. **6***f*, pad inner angles **151** to **153** of a first pentagonal friction lining pad **111** are indicated. The angle **151** is ninety degrees. The angle **152** is 110.9 degrees. The angle **153** is 145.7 degrees.
(52) In FIG. **6***g*, rounding radii **154** to **158** are indicated. The rounding radii **154**, **155**, **157** **158** are one millimeter. The rounding radius **156** is two millimeters.
(53) In FIG. **6***h*, double arrows **159**, **160** indicate a width and a height of the first pentagonal friction lining pad **111**. The width **159** is sixteen millimeters. The height **160** is 6.2 millimeters.
(54) In FIG. **6***i*, double arrows **161**, **162** indicate widths of radial grooves between the first pentagonal friction lining pads **111**. The widths **161**, **162** are two millimeters. The width **164** of the first friction lining pad arranged therebetween is sixteen millimeters. A width of a radial groove between two adjacent second friction lining pads **112** is indicated by dimension arrows at **163** and is one millimeter.
(55) FIG. **7***a* shows a groove angle **7** between two embossed grooves **165**, **166** of a V-shaped double groove. A first radial groove **167** has a width **8**. The embossed groove **166** has a width **9**. The groove angle **7** is between twenty and thirty degrees. The groove angle **7** may be 24.3 degrees. The width **9** of the embossed groove **166** is smaller than the width **8** of the radial groove **167**. An embossing depth of the embossed groove **166** at most corresponds of one half of the lining thickness of the respective friction lining pad.
(56) In FIG. **7***b*, it is indicated at **10** that the first pentagonal friction lining pads **111** can also be segmented into three parts. Corresponding segmentation grooves are designated with **168** and **169**.
(57) FIG. **7***c* shows groove widths **171** and **172**. The groove width **171** of the corresponding radial groove is two millimeters. The groove width **172** of the corresponding embossed groove in the first friction lining pad **111** is 1.2 millimeters. The groove angle **173** is 24.3 degrees. The V-shaped double groove can be embossed, milled or punched.
(58) In FIG. **8***a*, pad inner angles **1** are shown on one of the second pentagonal friction lining pads **112**. In FIG. **8***b*, rounding radii **2** of the second pentagonal friction lining pads **112** are shown. In FIG. **8***c*, a width **3** and a height **4** of the second pentagonal friction lining pads **112** are shown.
(59) The pad inner angles **1** are between ninety and one hundred and fifty degrees. The rounding

radii **2** are greater than or equal to one millimeter. A width **3** to height **4** ratio of the second friction lining pads **112** is between one and three, e.g., **2.33**. The pad inner angle **181** in FIG. **8***d* is 107.1 degrees. The pad inner angle **182** is 90.7 degrees. The pad inner angle **183** is 138.2 degrees.

(60) The width **184** in FIG. **8***e* is sixteen millimeters. The height **185** in FIG. **8***e* is 7.2 millimeters.

(61) In FIG. **8***f*, the rounding radii **186** to **189** are one millimeter. The rounding radius **190** is two millimeters.

(62) In FIG. **9***a*, it is indicated that two second friction lining pads **142**, **143** are integrally connected to each other in each case. FIG. **9***b* shows that three second friction lining pads **142** to **144** are integrally connected to each other in each case. FIG. **9***c* shows that four friction lining pads **142** to **145** are integrally connected to each other in each case. The integrally connected second friction lining pads are each divided only by embossed grooves **126**.

(63) In FIG. **9***d*, a groove width **191** of a second radial groove **122** designed as a segmentation groove is 1.8 millimeters. A groove width **192** of the embossed groove **126** is 1.2 millimeters.

(64) In FIG. **10***a*, a looped arrow indicates a schematic cooling oil flow in the closed state of the friction system. In a region **101**, the cooling oil flow can be maintained in the friction system due to the low segmented inner ring or due to a reduced flow cross-section on the inner ring and the curved tangential groove. Outflow to the outside and inside can be minimized.

(65) In FIG. **10***b*, a schematic oil removal in the open and closed state is illustrated by arrows. In a region **102**, there is an improvement in the friction value build-up due to effective oil removal in the lubrication gap in the segmentation as well as in the over-embossing. Thus, a smooth closing of the friction system can be realized during actuation.

(66) In the regions **103**, a diffuser effect can be realized by the groove cross-sectional area increasing from the inside to the outside. This can improve the air intake. In addition, drag losses can be reduced.

(67) A coordinated segmentation of the inner ring can reduce indirect oiling from the inside. This additionally allows air to enter the lubrication gap.

REFERENCE NUMERALS

(68) **1** Pad inner angle **2** Rounding radii **3** Widths **4** Heights **5** Outer flow cross-section **6** Inner flow cross-section **7** Groove angle **8** Width **9** Width **10** Pad variant **20** Multiple disc brake **21** Lubrication system **22** Lubrication system **23** Lubrication system **24** Arrow **25** Double arrow **26** Arrow (external oiling) **27** Inner multiple disc carrier **28** Friction disc **29** Steel disc **30** Outer multiple disc carrier **31** X-axis **32** Y-axis **33** Rotational axis **34** Frictional surface **35** Internal toothing **36** Circle **37** Circle **38** Speed drop **39** Speed increase **40** Rectangle **41** X-axis **42** Y-axis **43** Curve **44** Temperature distribution **45** Section **46** Section **47** Section **48** Circle **49** Circle **50** Shift **61** Friction disc **62** Groove pattern **63** Groove pattern **64** Groove pattern **65** Groove pattern **66** Groove pattern **67** Groove pattern **68** Groove pattern **69** Groove pattern **70** Linear course **71** Point **72** Maximum **73** Drop **101** Region **102** Region **103** Regions **111** First pentagonal friction lining pads **112** Second pentagonal friction lining pads **113** Carrier disc **115** Groove pattern **120** Circumferential groove **121** First radial groove **122** Second radial groove **123** Double groove **124** Embossed groove **125** Embossed groove **126** Embossed groove **127** Segmentation groove **131** First friction lining pad **132** First friction lining pad **133** First friction lining pad **134** First friction lining pad **141** Second friction lining pad **142** Second friction lining pad **143** Second friction lining pad **144** Second friction lining pad **145** Second friction lining pad **151** Pad inner angle **152** Pad inner angle **153** Pad inner angle **154** Rounding radius **155** Rounding radius **156** Rounding radius **157** Rounding radius **158** Rounding radius **159** Width **160** Height **161** Groove width **162** Groove width **163** Groove width **164** Width **165** Embossed groove **166** Embossed groove **167** Radial groove **168** Segmentation groove **169** Segmentation groove **171** Groove width **172** Groove width **173** Groove angle **175** First row of pads **176** Second row of pads **181** Pad inner angle **182** Pad inner angle **183** Pad inner angle **184** Width **185** Height **186** Rounding radius **187** Rounding radius **188** Rounding radius **189** Rounding radius **190** Rounding radius **191** Groove width **192** Groove width

Claims

1. A groove pattern for an annular wet friction system having external lubrication, comprising a frictional surface having a circumferential groove extending circumferentially in a zigzag or undulating manner, wherein: the circumferential groove is arranged, in a radial direction, between first pentagonal friction lining pads arranged radially outwards in a first row of pads and second pentagonal friction lining pads arranged radially inwards in a second row of pads; the first and second pentagonal friction lining pads have a rectangular geometry with an immediately adjacent triangular geometry; tips of the triangular geometry of the first pentagonal friction lining pads are directed radially inwards and tips of the triangular geometry of the second pentagonal friction lining pads are directed radially outwards; first radial grooves are arranged in a circumferential direction between the first pentagonal friction lining pads and second radial grooves are arranged in the circumferential direction between the second pentagonal friction lining pads; the tips of the triangular geometry of the first pentagonal friction lining pads face the second radial grooves and the tips of the triangular geometry of the second pentagonal friction lining pads face the first radial grooves; the first pentagonal friction lining pads have a V-shaped double groove comprising two stamped grooves spaced apart in the circumferential direction with a distance therebetween decreasing radially outwards; an angle between the two stamped grooves is between twenty (20) degrees and thirty (30) degrees; and the two stamped grooves each extend uninterruptedly from an outer edge of the first pentagonal friction lining pad towards an outer edge of a respective second pentagonal friction lining pad and open into the circumferential groove.
2. The groove pattern according to claim 1, wherein two second pentagonal friction lining pads are integrally connected to each other and are divided only by an embossed groove.
3. The groove pattern according to claim 1, wherein pad inner angles in pad corners of the first and second pentagonal friction lining pads are between ninety and one hundred and fifty degrees.
4. The groove pattern according to claim 1, wherein the first and second pentagonal friction lining pads have widths and heights that have a width to height ratio that is greater than one and less than three.
5. The groove pattern according to claim 1, wherein a radial flow cross-section between the first pentagonal friction lining pads is larger than a radial flow cross-section between the second pentagonal friction lining pads.